DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2019/2020

FIRST YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE

CCS2302: DISTRIBBUTED SYSTEMS

DATE: 11TH DECEMBER, 2019 TIME: 08.30-10.30AM

Instructions to candidates:

Answer Question 1 and any other 3 Questions. Every question should begin on a FRESH page and should be well labeled

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Question 1 COMPULSORY (30 Marks)

a) Describe the following terms as used in distributed systems (10 marks)

(i) Openness
(ii) Scalable system
(iii) Binding by identity
(iv) Binding by value
(v) Unattached resources

b) Describe the two main strict consistency techniques as used in replication management (10 marks)

c) Briefly describe the two clock sychronization algarithms (use appropriate diagrams) ***(10 marks)***

Question 2 (20 marks)

a) The DNS is one of the largest naming services in use today. Write short notes on the following in as far as the following is concerned

(i) DNS name space (5 marks)

(ii) DNS Implementation (5 marks)

b) Describe two methods you would use in locating mobile entities (10 marks)

Question 4 (20 marks)

a) Briefly describe how PGP is used in securing email communication (2 marks)

b) Describe two approches you would use in securing an electronic payment system (6 marks)

c) The WWW is the best example of a widely distributed system. Discuss WWW with respect to: (12 marks)

(i) How it is organised

(ii) HTML

(iii) HTTP

(iv) Processes

(v) Server clusters

(vi) Naming